

CURRICULUM-VITAE

Name : Dr. Karuna Nidhan Singh
Discipline : Agril. Engg. (Soil and Water Engineering)
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Carrier Objectives:

To contribute agricultural innovative knowledge among the students for development of their skills and growth of institute.

Academic Qualifications:

Level	Degree	Board/Institute/University	Year	Subjects/Major Field	Division with aggregate %	Rank/Medal
Doctorate	Ph.D.	IGKV, Raipur, C.G.	2022	Soil and Water Engg.	1 st (79.60)	—
Master	M.Tech	SHIATS, Prayagraj, U.P.	2014	Soil and Water Cons. Engg.	1 st with honors (9.3/10)	2 nd (Silver Medal)
Graduation	B.Tech.	M.G.C.G.V. Chitrakoot, Satna, M.P.	2011	Agricultural Engg.	1 st (66.00)	—
HSC	12 th	U.P. Board	2006	Mathematics	2 nd (54.60)	—
SSC	10 th	U.P. Board	2004	Science	1 st (70.83)	—

Project Details:

- ❖ Impact Assessment of Low Cost Rainwater Harvesting Structures on Watershed Hydrology in Bundelkhand Region of Central India during **Ph.D.** degree programme under the supervision of **Dr. Ramesh Singh**, Head, ICRISAT Development Centre, Patancheru, Hyderabad, Telangana (Former Principal Scientist, ICAR-CAFRI, Jhansi, U.P.) and **Dr. Dhiraj Khalkho**, Principal Scientist/Professor, IGKV, Raipur, Chhattisgarh.
- ❖ Development of Single Rainstorm Erosivity Model for Chitrakoot District during **M.Tech.** degree programme under the supervision of **Dr. R. K. Isaac**, Professor, SHIATS, Allahabad, Uttar Pradesh.
- ❖ Impact of Recharge Structure of Dug Well and Blast Well for Increasing Water Level at Different Regions during **B.Tech.** degree programme under the supervision of **Er. Ashwani Duggal**, Assistant Professor, MGCGV, Chitrakoot, Satna, Madhya Pradesh.

Experience in Academic /Administration:

A. Employment Record

- ❖ Currently working as **Assistant Professor** in School of Agriculture, ITM University, Gwalior, M.P from April, 2023.
- ❖ Working also as **Training Co-ordinator** in School of Agriculture, ITM University, Gwalior, M.P.
- ❖ Working also as **Chief of the Department (CoD)** in School of Agriculture, ITM University, Gwalior, M.P.
- ❖ Working also as **Meteorological Observatory In-charge** in School of Agriculture, ITM University, Gwalior, M.P.
- ❖ **Three years** teaching experience as **Assistant Professor** in Faculty of Agricultural Engineering & Technology, AKS University, Satna, M.P.
- ❖ Worked also as **Head of Department (I/C)** in Faculty of Agricultural Engineering & Technology, AKS University, Satna, M.P.

B. Institute/University Service Function

- ❖ Working also as **Assistant Centre Superintendent (Exam)**, ITM University, Gwalior, M.P.
- ❖ Working also as **Counsellor** in the counselling cell, ITM University, Gwalior, M.P.
- ❖ Working also as **Member in NAAC Criteria-I**, ITM University, Gwalior, M.P.
- ❖ Working also as Member in **Flying Squad (Exam Cell)**, ITM University, Gwalior, M.P.
- ❖ Worked as **NRM expert** in ICRISAT, ICRISAT Development Centre, Jhansi, U.P.
- ❖ Three years worked as **Counsellor** in the counselling cell, A.K.S. University, Satna, M.P.

C. Research Guidance as Major Guide/Advisor

- ❖ Guided four B.Tech level students as Major guide/advisor during the session of 2023-24 in ITM University, Gwalior, M.P.
- ❖ Guided three B.Tech level students as Major guide/advisor during the session of 2016-17 in A.K.S. University, Satna, M.P.
- ❖ Guided four B.Tech level students as Major guide/advisor during the session of 2017-18 in A.K.S. University, Satna, M.P.

D. Membership of Organization/Association/Societies

- ❖ Member of All India Agricultural Students Association (AIASA), New Delhi with membership- AIASA/IGKVV/OL/LM-331/2019.
- ❖ Life member of Society of Extension Education (SEE), Agra with Life membership-UP/St 660/2020.
- ❖ Life member of Indian Society of Agricultural Engineers (ISAE), New Delhi with Life membership-LM/ISAE/12026/2020.
- ❖ Life member of International Association for Water Environment Energy and Society (IAWEES), IIT, Bombay with Life Membership-M-5367/2020.
- ❖ Life member of Indian Society of Remote Sensing (ISRS), Dehradun with Life membership-LM/ISRS/5600/2020.
- ❖ Life member of Indian Water Resources Society (IWRS), IIT, Roorkee with Life membership-LM-2020-7874/2020.

- ❖ Life member of Indian Association of Hydrologists (IAH), NIH, Roorkee with Life membership-LM/IAH/2034/2021.
- ❖ Life member of International Society for Development and Sustainability (ISDS), Japan with membership ID-M21052101.
- ❖ Life member of Agricultural Technology Development Society (ATDS), Ghaziabad (U.P.) with life member certificate number-2315/2021.

E. Institutional/Professional Trainings

1. National Trainings

- ❖ Four weeks Institutional/Professional Training organized by Central Farm Machinery Training and Testing Institute, Budhni, M.P.
- ❖ Two weeks training on Hydrological Modelling using QGIS Interface with QSWAT organized by National Institute of Hydrology, Roorkee, U.K.
- ❖ Ten days Skill Development course on Hands on Crop Modeling (DSSAT) organized by Agromet Field unit, Srinagar in collaboration with NAHEP, SKUAST, Kashmir.
- ❖ Seven Days Online Capacity Building Program on Hydro-Meteorological Disaster: Techniques of Measuring, Monitoring, and Management of Floods organized by Shaheed Bhagat Singh College, University of Delhi, New Delhi.
- ❖ Two days national training on Creating Awareness about Intellectual Property Rights Protection for the Start-Ups and Entrepreneur in Agri-horti Sectors organized by the College of Horticulture and Forestry, Central Agricultural University (Imphal), Pasighat, Arunachal Pradesh.
- ❖ Four days virtual training programme on Recent Trends in Cultivation, Processing and Value Addition of Few Important Medicinal and Aromatic Plants (MAPs) under the NAHEP-CAAST sub-project organized by Navsari Agricultural University, Navsari.
- ❖ Ten days online training programme on Rainfall Data Analysis using different Packages of R-Software organized by KSCSTE-CWRDM, Kozhikode, Kerala.
- ❖ Three days online Training Course on Basics of Remote Sensing and GIS for Management of Soil and Water Resources organized by Department of Natural Resources Management, College of Forestry, Banda University of Agriculture & Technology, Banda.
- ❖ Three days online Training Course on Soil Health Management Concepts and Practices organized by ANRCM, Lucknow.

2. International Trainings

- ❖ One-week International Training Program on Water Resources Modelling organized by CAAST-CSAWM, Mahatma Phule Krishi Vidyapeeth, Rahuri, Maharashtra.

F. Conference/Workshop/Seminar/Symposium Attended

1. National

- ❖ Participated in **National Conference** on water demand, sustainability and future organized by River Water User Association and Institution of Engineers in Motilal Nehru National Institute of Technology (MNNIT), Allahabad, U.P. during May 03-04, 2014.
- ❖ Participated in Career Counselling **Workshop** on higher education, scholarships and job opportunity in USA and other countries organized by University Placement Cell in Indira Gandhi Krishi Vishwavidyalaya, Raipur, C.G. on March 13th, 2018.

- ❖ Participated in **National Conference** on Resource Conservation for Soil Security and Jalshakti: Farmers Perspective in Bundelkhand organized by Indian Association of Soil and Water Conservationists (IASWC), Dehradun in ICAR-Indian Institute of Soil and Water Conservation & Research Center, Ditiya, M.P. during Feb 03-05, 2020.
- ❖ Participated in **National Webinar** on Impact of Population Pressure on Natural Resources and Environment organized by Academy of Natural Resource Conservation and Management (ANRCM) in Lucknow, Uttar Pradesh, India on July 11th, 2020.
- ❖ Participated in **National Webinar** on On-farm Skill Development Through Soil and Water Conservation Technologies: A Buffer Against Production Skill Risk in the Face of Climate organized by Rajmata Vijayaraje Scindia Krishi Vishwavidyalaya (RVSKV) in Gwalior, Madhya Pradesh on July 25th, 2020.
- ❖ Participated in **National Web-Conference** on Technological Approaches for Resource Conservation and Management for Environmental Sustainability organized by Academy of Natural Resource Conservation and Management (ANRCM) in Lucknow, Uttar Pradesh, India during August 16-17, 2020.
- ❖ Participated in **National Webinar** on Hydro informatics for Smart Water Management in Agriculture organized by CAE, RPCAU, Pusa; NIH, Roorkee; Deptt. of WRDM, IIT Roorkee & Div. of Agril. Engg. in ICAR-IARI, New Delhi on October 20th, 2020.
- ❖ Participated in **National Web Conference** on Sustainable Fodder Production for Improving the Livelihood of Small and Marginal Farmers organized by Bihar Agricultural University in Sabour, Bhagalpur, Bihar on November 26th, 2020.
- ❖ Participated in **National e-Posters Olympiad** on Soils, Biomes and Resilience to Climate Change organized by Soil Conservation Society of India, New Delhi in SCSi, New Delhi during December 4-5, 2020.
- ❖ Participated in **National Essay Competition** on Adoption of Digital Technologies by Indian Farmers during and after COVID-19 Pandemic organized by Vasantrao Naik Marathwada Krishi Vidyapeeth, Prabhani in Prabhani, Maharashtra on December 23th, 2020.
- ❖ Participated in the 29th **National Web Conference** on Sustainable Soil and Water Management for Bio-diversity Conservation, Food Security and Climate Resilience organized by Soil Conservation Society of India, New Delhi in SCSi, New Delhi during December 29-30, 2020.
- ❖ Participated in **National Seminar** on Restore Our Earth organized by IIM, Sirmaur on April 22nd, 2021.
- ❖ Participated in the **National Webinar** on Our Solutions are in Nature organized by Directorate of Research, Maharana Pratap University of Agriculture and Technology, Udaipur, Rajasthan on May 22nd, 2021.
- ❖ Participated in the **Webinar** on Digital Farming in irrigation and Fertigation organized by Albedo Foundation on May 23rd, 2021.
- ❖ Participated in the **National Webinar** on Organic Animal Production: Opportunities and Strategies in India organized by Niche Area of Excellence on Organic Poultry Production, Directorate of Research, Maharana Pratap University of Agriculture and Technology, Udaipur, Rajasthan on June 1st, 2021.
- ❖ Participated in the **National Webinar** on Water Management: Opportunities & Challenges organized by Department of Soil and Water Engineering, College of Technology and Engineering, MPUAT, Udaipur, Rajasthan, India on June 3rd, 2021.

- ❖ Participated in the **National Workshop** on Water budgeting: An Approach for Sustainable Water Resource Management in Rajasthan, organized by Directorate of Research, MPUAT, Udaipur, Rajasthan, India on June 21st, 2021.
- ❖ Participated in the **National Webinar** on Sustainable Strategies for Enhancing Productivity & Profitability in Dryland Agriculture organized by AICRP on Dryland Agriculture, Dryland Farming Research Station, Arjia, Bhilwara, Directorate of Research, Maharana Pratap University of Agriculture and Technology, Udaipur (Rajasthan) on June 29th, 2021.
- ❖ Participated in the **National Webinar** on Conservation Agriculture in India : Myths, Realities and Way Forward organized by Department of Agronomy, Rajasthan College of Agriculture, Maharana Pratap University of Agriculture and Technology, Udaipur (Rajasthan) on February 17th, 2022.
- ❖ Participated in the **National Webinar** on Quit India Movement to Food Secured India organized by Maharana Pratap University of Agriculture and Technology, Udaipur, Rajasthan on August 08th, 2022.

2. International

- ❖ Participated and presented paper in the **International Web-Conference** on Resource Management and Biodiversity Conservation to Achieve Sustainable Development Goals organized by Academy of Natural Resource Conservation and Management (ANRCM) in Lucknow, Uttar Pradesh, India during September 11-12, 2020.
- ❖ Participated in **2nd Asian Web Conference** on Manage Hill Resources and Diversities for Zero Hunger and Climate Resilience organized by Soil Conservation Society of India, New Delhi in SCSi, New Delhi during February 12-13, 2021.
- ❖ Participated in **International Conference** on Soil and Water Resource Management organized by College of Technology and Engineering, MPUAT, Udaipur Rajasthan in Udaipur, Rajasthan, India during February 26-27, 2021.
- ❖ Participated in the **4th International Conference** on Global Approaches in Natural resource Management for Climate smart agriculture (GNRSA-2020) during Pandemic era of COVID-19 Organized by Agricultural Technology Development Society (ATDS), Ghaziabad (U.P.), India during February 26-28, 2021.
- ❖ Participated in **International Conference** on Sustainable Water Resources Development and Management organized by Govt. College of Engineering, Aurangabad (Maharashtra) in Aurangabad, Maharashtra, India during March 8-9, 2021.
- ❖ Participated in the two days **International Conference** on Recent Challenges and Opportunities in Engineering organized by the CTAE, Udaipur during March 13-14, 2021.
- ❖ Participated in **International Conference** on Emerging Techniques in Food Processing (ETFP) organized by Department of Food Processing Technology, Ghani Khan Choudhury Institute of Engineering and Technology Malda during March 25-26, 2021.
- ❖ Participated and presented paper in the **International Conference** on the Future Challenges and Sustainable Development Goals: Science to Policy Frame work organized by Shaheed Bhagat Singh college, University of Delhi, India during April 16-18, 2021.
- ❖ Participated in an online **International Webinar** on Sustainable Groundwater Management: Current Challenges Around the World, Organized by Department of Soil and Water Engineering, CTAE, MPUAT, Udaipur, Rajasthan, India on July 30th, 2021.
- ❖ Participated in **3rd International Web-conference** on Natural Resource Management for Global Food security and Sustainable Goals Organized by the Academy of Natural

Resource Conservation and Management (ANRCM) in Lucknow, Uttar Pradesh, India during December 02-03, 2022.

- ❖ Participated in the **International conference** of Next-Generation Agriculture for Sustainable Environment and One Health organized by Swami Vivekanand Subharti University, Meerut, India during 09-11 February, 2024.

G. Poster/Paper Presentations

- ❖ **K. N. Singh** and R. Singh presented **Paper** on entitled Study of livelihood Security by Adoption of Cost Effective NRM Through Agroforestry in SAT Region, Central India organized by Academy of Natural Resource Conservation and Management (ANRCM) in Lucknow, Uttar Pradesh, India during International Web-Conference during September 11-12, 2020.
- ❖ **K N Singh**, R Singh, D Khalkho and M P Tripathi presented a **Paper** on entitled RS and GIS Based Landscaping for Efficient Management of Hill Resources and Watershed Development for Sustainable food and Livelihood Security-A Case Study in Bundelkhand Region, Central India organized by SCSi, New Delhi during the 2nd Asian Web Conference during 12-13 Feb, 2021.
- ❖ **K N Singh** and R Singh presented a **Poster** on entitled RS and GIS Based Impact Assessment of Soil and Water Conservation Practices on food sustainability and Livelihood Security-A Case Study in Bundelkhand Region, Central India organized by MPUAT, Udaipur, Rajasthan, India during International Conference during 26-27 Feb, 2021.
- ❖ **K. N. Singh**, Liansunpui, R. Singh and D. Khalkho presented **Paper** on entitled RS and GIS based Soil and Water Conservation Approaches-Today's needs and their solutions for livelihood Security-a case study in Bundelkhand Region, Central India organized by ATDS, Ghazibad, Uttar Pradesh, India during February 26-28, 2021.
- ❖ **K N Singh**, R Singh, D Khalkho and M P Tripathi presented a **Paper** on entitled RS and GIS based landscaping planning, designs, conservation practices for sustainable water management along with food and livelihood security - A case study in GKD watershed, Central India organized by Shaheed Bhagat Singh college, University of Delhi, India during April 16-18, 2021.
- ❖ **K. N. Singh** (2024). Implementation of Water Harvesting Structures through RS and GIS for Sustainable Ecosystem Services in the International conference of Impact of Climate Changes on Global Food, Livestock, Livelihood and Environmental Security: Advanced organized by NAU, Navsari and NADCL, Baramulla, in-collaboration with ICAR-NAHEP and others during 28 to 30 December 2023.
- ❖ **K. N. Singh** (2024). Watershed Management through RS and GIS – A Success Approach for Sustainable Agriculture and Ecosystem in the International conference of Next-Generation Agriculture for Sustainable Environment and One Health organized by Swami Vivekanand Subharti University, Meerut, India during 09-11 February, 2024.

Publications:

A. Patents

- ❖ A design **patent** is granted with patent no. 403389-001 on titled **Double Ring Pressure Infiltrimeter** in 2024.
- ❖ A **patent** is published with patent no. 202321070573 on titled **AI based drip Irrigation system** in 2023.

B. Research Papers

- ❖ **Singh, K.N.**, Dalai, A., Kaiwart, M.P., Mohanty, R.R. and Kumar, B. 2018. Prediction of rainfall of Allahabad district by the development of autoregressive time series model. *Int. J. Curr. Microbiol. App. Sci.*, Vol. 7(4): 1516-1522.
- ❖ **Singh, K.N.**, Dalai, A. and Gautam, A.K. 2018. Design of sprinkler irrigation system for 5 hectare land at Sohawal block in Satna district of Madhya Pradesh. *International Journal of Chemical Studies*, Vol. 6(3): 3228-3232.
- ❖ **Singh, K.N.**, Dalai, A., Mohanty, R.R. and Isaac, R.K. 2018. Development of single rain storm erosivity models in central plateau and hill zones for Chitrakoot district. *Journal of Pharmacognosy and Phytochemistry*, Vol. 7(2): 2961-2965.
- ❖ Mohanty, R.R., Dalai, A., **Singh, K.N.**, Das, A. and Dalai, B. 2018. New approach in rain water harvesting through rooftop water management technique-case study. *International Journal of Agriculture Sciences*, Vol. 10(8): 5815–5816.
- ❖ **Singh, K.N.**, Singh, R., Khalkho, D. and Singh, A.K. 2019. Estimation of soil erosion caused by surface runoff using DAS operating technique in Bundelkhand region. *International journal of Agriculture Sciences*, Vol. 11(7): 8183–8185.
- ❖ **Singh, K.N.**, Singh, R., Khalkho, D., Singh, A.K. and Tripathi, M.P. 2019. Ground water recharge potential using different analytical approaches in Parasail-Sindh watershed of Babina block, Bundelkhand region-a case study. *Research Journal of Chemical and Environmental Sciences*, Vol. 7 (2): 58-62.
- ❖ **Singh, K.N.**, Singh, R., Liansangpuii and Khalkho, D. 2019. Comparative study of micronutrient cum fertilizers on crop production and benefit-cost ratio in Babina block, Bundelkhand region. *International Journal of Chemical Studies*, Vol. 7(5): 1914-1916.
- ❖ **Singh, K.N.**, Singh, R., Khalkho, D., Singh, A.K., Dahate, Amit and Tripathi, M.P. 2019. Environmental benefits by implementation of agroforestry systems for Garhkundar-Dabar watershed central India-a case study. *Research Journal of Chemical and Environmental Sciences*, Vol. 7 (2): 08-13.
- ❖ **Singh, K.N.**, Patil, B. and Dalai, A. 2021. Study of Water Table Behavior by Using Recharge Unit in Babina Block, Bundelkhand Region, Central India. *Biological Forum – An International Journal*, 13(1): 713-720.
- ❖ **Singh, K. N.**, Liansangpuii, Dalai, A., Dahate, A. and Rajawat, B. S. 2021. Hydrological Approach through V-Notch Weir for Discharge Measurement in an Agricultural Land and Strategies to Mitigate Land Degradation. *Res. J. Chem. Env. Sci.* Vol. 9 (3): 01-03.
- ❖ **Singh, K.N.**, Khalkho, D., Singh, R. and Tripathi, M.P. 2021. Enhancing Groundwater Recharge by Implementations of various Agricultural Water Intervention Structures-A Case Study in Bundelkhand Region, Central India. *Biological Forum – An International Journal*, 13(3): 118-122.
- ❖ Liansangpuii, **K. N. Singh**, R. Singh, R. M. Singh and D. Khalkho 2022. Statistical and temporal trends analysis of rainfall in Bundelkhand region, Central India. *Environment Conservation Journal* 23 (1&2): 131-142.
- ❖ **Singh, K.N.**, Singh, R., Khalkho, D. and Tripathi, M.P. (2022). Morphometric analysis for sustainable development of natural resource management by using remote sensing and GIS techniques in Tikamgarh District, M.P., Central India. *Environment Conservation Journal*, 23(1&2): 417-425.
- ❖ **Singh, K.N.**, Rajawat, B.S., Dahate, A. and Liansangpuii 2023. Development of Watershed Programme for Environment and Agriculture Sustainability in SAT Region, Central India. *Agricultural Reviews*. Volume 44 Issue 3: 415-419.

- ❖ Liansangpuii, Ramesh Singh, R. M. Singh, **K. N. Singh** and S. K. Kar (2023). Hydrological assessment of Haveli-based traditional water harvesting system for the Bundelkhand Region, Uttar Pradesh, India. *Current Science*, Vol. 125, No. 1.
- ❖ **Singh, K.N.**, Khalkho, D., Singh, R. and Tripathi, M.P. 2024. Impact of Economic Assessment of Crop Production along with Agroforestry Adoption in Bundelkhand Region. *Agricultural Reviews*. Volume 45 Issue 1: 177-181.
- ❖ **K. N. Singh**, Vipin Kumar and Sandip Kumar Gautam (2024). Environmental benefits by adopting agroforestry systems: A case study in the Bundelkhand region of Central India. *International Journal of Advanced Biochemistry Research*; SP-8(4): 461-464.
- ❖ **K. N. Singh**, Sandip Kumar Gautam and Vipin Kumar (2024). Investigating groundwater recharge potential through various analytical methodologies: A case study. *International Journal of Research in Agronomy*; SP-7(4): 287-290.
- ❖ Jaya Bindal, Vipin Kumar, **K. N. Singh** and MR Kadivar (2024). Effect of zinc scheduling on yield and yield attributes of wheat Crop (*Triticum aestivum* L.): *International Journal of Research in Agronomy*; 7(2): 509-511.
- ❖ Liansangpuii, Ramesh Singh, R.M. Singh, **K. N. Singh**, S.K. Kar (2024). Impact assessment of rainwater management interventions on land use land cover changes in Parasai-Sindh watershed of Bundelkhand Region, Central India. *Ecological Engineering* 201, 107213.

C. Abstracts:

- ❖ **Singh, K.N.**, Dalai, A., Kaiwart, M.P. and Kumar, B. 2018. Development of Autoregressive Time Series Model for Prediction of Rainfall for Allahabad District. ISAE-2018/SWE/SP-19.
- ❖ **K. N. Singh**, R. Singh, D. Khalkho and M. P. Tripathi 2020. Study of Morphometric Parameters for Best Water Conservation Practices in GKD watershed, Central India by Using Remote Sensing and GIS. www.anrcm.org, National Web-Conference held on 16-17 August, 2020, Page no. 211 in E-book.
- ❖ **K. N. Singh** and R. Singh 2020. Study of livelihood Security by Adoption of Cost Effective NRM Through Agroforestry in SAT Region, Central India. www.anrcm.org, International Web-Conference held on 11-12 September, 2020, Page no. 371 in E-book.
- ❖ **K N Singh**, R Singh, D Khalkho and M P Tripathi 2021. RS and GIS Based Landscaping for Efficient Management of Hill Resources and Watershed Development for Sustainable food and Livelihood Security-A Case Study in Bundelkhand Region, Central India. <http://www.scsi.org.in>, 2nd Asian Web Conference held on 12-13 February, 2021, Page no. 33 in E-book.
- ❖ **Singh K N**, Singh R 2021. Recent Interventions Used for Crop Improvement for Sustainable Food and Livelihood Security -A Case Study for Present and Future Scenario in Bundelkhand Region, Central India. ICRAAS 2021 eBook, International Conference held on 16-17 March, 2021, Page no. 44 in E-book.
- ❖ **K. N. Singh** and R Singh 2021. Essential prerequisites of food production and availability for food security, safety and sustainability without associated ecological harm-a case study in central India. Souvenir International Conference Emerging Technologies in Food Processing (ETFP-2021), 25-26 March, 2021, Page no. 151.
- ❖ **K. N. Singh**, Liansangpuii, R. Singh and D. Khalkho 2021. RS and GIS Based Soil and Water Conservation Approaches-Today's Needs and Their Solutions for Livelihood Security-A Case Study in Bundelkhand Region, Central India. Souvenir 4th International

Conference on Global Approaches in Natural resource Management for Climate smart agriculture (GNRSA-2020) during Pandemic Era of COVID-19, 26-28 February, 2021, Page no. 464.

- ❖ Liansangpuii, **K. N. Singh**, R. Singh and R. M. Singh 2021. Enhancing Groundwater Resources Availability through Implementation of Agricultural Water Intervention Structures - A Case Study. Souvenir 4th International Conference on Global Approaches in Natural resource Management for Climate smart agriculture (GNRSA-2020) during Pandemic Era of COVID-19, 26-28 February, 2021, Page no. 464.
- ❖ **K N Singh**, R Singh, D Khalkho and M P Tripathi 2021. RS and GIS Precision Tools for Smart Management of Natural Resources -A Case Study in GKD Watershed, Bundelkhand Region, Central India. www.anrcm.org, 2nd International Web-Conference on Smart Agriculture for Resource Conservation and Ecological Stability, October 29-31, 2021, Page no. 171.
- ❖ **K. N. Singh** and R. Singh 2022. Strategy of Natural Resource Conservation and Management for Sustainable Food Supply and Livelihood Security-A Case Study in Semi-Arid Tropics, Bundelkhand Region, Central India. 3rd International Web-Conference on Natural Resource Management for Global Food Security and Sustainable Development Goals December 02-03, 2022, Page no. 19.

D. Books:

- ❖ **K N Singh**, Anil Kumar, Sabnam, Diksha and Sandip Kumar Gautam (2023). Reference Book of Dimensions of Agriculture, ISBN-9781804339435, Rubicon Publication, England; 186.
- ❖ **K N Singh** and Madhukar Patel (2024). Subject book of Pump Engineering and Hydraulic Control. ISBN-9788119820566, ChyrenPublication, India; 160.
- ❖ **K N Singh**, Vipin Kumar and Rahul I Navsare (2024). Reference Book of Soil and Water Conservation Engg. ISBN-9789361352188, Akinik Publication, New delhi, India; 97.

E. Book Chapters:

- ❖ **K. N. Singh**, Deepali Suryawanshi, Vipin Kumar and Madhukar Patel (2024). Strategies, Implementation and Research Issues during Development of Sustainable Watershed Management for Livelihood Security. Organic Farming Practices and Sustainable Agriculture, Volume – 10, ISBN: 978-93-6135-651-3; 15-28.
- ❖ **K. N. Singh**, Vipin Kumar, Deepali Suryawanshi and Madhukar Patel (2024). Micro Irrigation Systems Adoption, challenges and Implementation for Sustainable Agriculture. Organic Farming Practices and Sustainable Agriculture, Volume – 10, ISBN: 978-93-6135-651-3; 29-42.
- ❖ **K. N. Singh**, Vipin Kumar, Deepali Suryawanshi and Sandip Kumar Gautam (2024). Agricultural Practices for Sustainable Crop Production. Emerging Trends in Agriculture Sciences, Volume – 16, ISBN: 978-93-5834-376-2, 131-144.
- ❖ Deepali Suryawanshi, **K.N. Singh** and Phool Singh Hindoriya (2023). Biodiversity and Bio Resources-Exploring Nature's Riches. Emerging Trends in Agriculture Sciences, Volume – 15, ISBN: 978-93-5834-677-0, 117-133.
- ❖ **K N Singh**, Vipin Kumar and Sandip Kumar Gautam (2024). Human and Animal Powered Based Surface Water and Groundwater Lifting Devices. Advances in Agricultural and Life Sciences, Volume – 14, ISBN: 978-3-96492-570-1, 103-119.

- ❖ **K N Singh**, Sandip Kumar Gautam and Vipin Kumar (2024). Agronomical Practices to Control Soil Erosion in Terms of Soil Conservation. *Advances in Agricultural and Life Sciences*, Volume – 14, ISBN: 978-3-96492-570-1, 121-135.
- ❖ **K. N. Singh**, Vipin Kumar and Deepali Suryawanshi (2023). Efficient Land and Water Management for Sustainable Development. *Capsule of Agriculture*, ISBN: 978-1-80433-959-6, 554-562.
- ❖ **K. N. Singh**, Deepali Suryawanshi and Vipin Kumar (2023). Promising Future in Soil and Water Conservation. *Capsule of Agriculture*, ISBN: 978-1-80433-959-6, 563-572.
- ❖ Deepali Suryawanshi, **K. N. Singh** and Vipin Kumar (2024). The Role of Agricultural Extension in Promoting Sustainable Farming Practices. *Scientific Inputs in Modern Indian Agriculture*, ISBN: 978-3-96492-574-9, 198-208.
- ❖ **K N Singh**, Vipin Kumar and Sandip Kumar Gautam (2024). Introductory of nano Patterning and their Methods. *Advances in Agriculture Sciences*, Volume -46, ISBN-9789361358647, 101-115.
- ❖ **K N Singh**, Vipin Kumar and Sandip Kumar Gautam (2024). Introductory of Nano Biology and Biomolecules, *Advances in Agriculture Sciences*, Volume -46, ISBN-9789361358647, 117-129.
- ❖ Suraj Kumar, Deepali Suryawanshi, **K. N. Singh** & Puneet Kumar (2024). Insect Ecology and Their Interaction with Biotic and Abiotic Components. *Ecology and Environment for Sustainable Development as per SDGs*, ISBN: 978-93-88361-61-3, 179-193.

F. Articles:

- ❖ **K. N. Singh**, S. K. Gautam and Vipin Kumar (2024). Agricultural engineering technology- An over view and their applications. *Agri Gate-An International Multidisciplinary e-Magazine*, ISSN-9788196558291;144-149.
- ❖ **K. N. Singh**, Vipin Kumar and S. K. Gautam (2024). Natural resource management –An important concern for 21st Century. *Agri Gate-An International Multidisciplinary e-Magazine*, ISSN-9788196558291; 209-294.
- ❖ **K. N. Singh**, Vipin Kumar and S. K. Gautam (2024). Water use Efficiency Assessment through Micro Irrigation Systems. *Eco farming e-Magazine for Agriculture and Allied Sciences*, ISSN-2583-0791; 111-115.
- ❖ **K. N. Singh**, Vipin Kumar and S. K. Gautam (2024). Introductory of Canals and their Importance, *Eco farming e-Magazine for Agriculture and Allied Sciences*, ISSN-2583-0791; 116-120.
- ❖ **K. N. Singh**, R. Singh, Vipin Kumar and Sandip Kumar Gautam (2023). Approaches for Soil and Water Conservation Practices in Central India. *Vigyan Varta an International E-Magazine for Science Enthusiasts*, ISSN-2582-9467; 157-159.
- ❖ **K. N. Singh**, R. Singh, Vipin Kumar and Sandip Kumar Gautam (2023). Crop Cultivation along with Agroforestry for Sustainable Development. *Vigyan Varta an International E-Magazine for Science Enthusiasts*, ISSN-2582-9467; 154-156.
- ❖ **K. N. Singh**, Vipin Kumar, S. K. Gautam and Deepali Suryawanshi (2024). Introductory of Nanoparticles and their Importance in Agriculture. *New Era Agriculture Magazine*, ISSN-2583-5173; 8-10.
- ❖ Deepak Singh, Vipin Kumar, **K. N. Singh** and Deepali Suryawanshi (2024). Future Aspects of Organic Farming in Agriculture. *New Era Agriculture Magazine*, ISSN-2583-5173; 15-18.

- ❖ **K. N. Singh**, Vipin Kumar, and Deepali Suryawanshi (2024). Role of Sprinkler Irrigation in Water Conservation. New Era Agriculture Magazine, ISSN-2583-5173; 22-25.
- ❖ निर्भय सिंह, करुणा निधान सिंह, व पन कुमार और दीपाली सूर्यवंशी (2024). पशु आहार में यूरिया का प्रभाव:लाभ और वचार. New Era Agriculture Magazine, ISSN-2583-5173; 1-4.

G. Invited Lectures

- ❖ Invited as a **Guest Speaker** for students and farmers by Society of HARWS, India invited by Pulkit Chaudhary, President on 28-07-2023.
- ❖ Invited as a **Guest Speaker** for farmers by KRIBHCO, Gwalior, MP invited by Abhishek Modi, Field Officer KRIBHCO on 01-02-2024.

Special Achievements:

- ❖ Selected in National Fellowship Award for Other Backward Classes (**NFO-OBC**) in 2018-19 (ID-201819-NFO-2018-19-OBC-UTT-72827) by UGC, Govt. of India.
- ❖ Secured **First** position for **Poster Presentation** during the Int. Conf. on Soil and Water Resource Management (ICSWRM 2021) during 26-27 February, 2021.
- ❖ Received **Young Scientist Award-2020** for Outstanding Contribution in the Field of Soil and Water Engineering on the occasion of 4th International Conference on Global Approaches in Natural resource Management for Climate smart agriculture (GNRSA-2020) during Pandemic Era of COVID-19 during 26-28 February, 2021.
- ❖ **Best Paper Presentation Award** in the International conference of Impact of Climate Changes on Global Food, Livestock, Livelihood and Environmental Security: Advanced organized by NAU, Navsari and NADCL, Baramulla, in-collaboration with ICAR-NAHEP and others during 28 to 30 December 2023.

Area of Interests:

- ❖ Reading and teaching of agricultural innovative technologies to develop skills and knowledge.
- ❖ Soil and water conservation measures/structures at watershed scale.
- ❖ Natural resource management along with agroforestry.

Computer/Software/Model Skills:

- ❖ Fundamental Knowledge of computer and M-S office.
- ❖ Agricultural Softwares and Modells viz. QGIS, ArcGIS, Erdas Imagine, SAGA, HOBOW are and SWAT, SWAT-Cup, SWAT-Interface respectively for designing of soil and water conservation measurements along with agroforestry to fulfill the objective of DFI and facilitate livelihood security.

Future Plan of Research Work:

- ❖ Contribute the efficient way of rainwater harvesting structures building systems resilience to students and local farm/village scale by enhancing groundwater availability.
- ❖ Provide the insights on spatial extend of the benefits of rainwater harvesting and guidelines on water harvesting protocol of any region.

- ❖ How to rejuvenate riverine ecosystem at various locations in drought prone regions by effective decentralized water harvesting as most of the rivers gets dry or holds very low flow soon after monsoon.

Strengths:

- ❖ Flexible and adaptive in nature
- ❖ Having Dedication towards work along with accuracy

Personal Details:

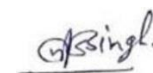
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Declaration:

I, hereby, declare that all the information is correct to best of my knowledge and belief.

Date: 25-02-2025

Place: Gwalior, M.P.



Karuna Nidhan Singh